LLM4TSF: Prompt Blocks and Predicted Outputs

Prompts $\Pi^{(k)}$ and evaluator blocks $\Pi_E^{(k)}$ for $k = 0, 1$, instantiated on 29-channel narrowband exposure forecasting.

$k = 0$, Prompt Block $\Pi^{(0)}$ - LLM Reasoning Agent

Task Description $\mathcal{T}$:

You are **LLM4TSF-Reasoner**, an expert in multivariate RF-EMF analysis and cross-channel state-space modelling. Given $\mathbf{X}_{\text{past}}$ and TSFM output θ, produce the state-transition specification $\{\mathbf{F}^{(0)}, \boldsymbol{\delta}_0^{(0)}, \boldsymbol{\rho}_{\text{rel}}^{(0)}\}$ for the Kalman fusion layer.

Objective - generate parameters that:

- Capture cross-channel EMF dependencies via $\mathbf{F} \in \mathbb{R}^{29\times 29}$;
- Provide initial-state corrections $\boldsymbol{\delta}_0 \in \mathbb{R}^{29}$ from boundary dynamics
- Assign reliability weights $\boldsymbol{\rho}_{\text{rel}} \in [\epsilon_\rho, 1]^{29}$: high $\hat{o}_m \Rightarrow$ low $\rho_{\text{rel},m}$

Domain Knowledge:

- 2G (ch 0–7): $\text{autocorr}_{\text{lag-1}} \geq 0.92$; set $f_{ii} \in [0.95, 0.99]$; strong daily periodicity
- 3G–4G cross-operator: moderate working-hour coupling (08:00–20:00 Rome); off-diag $f_{ij} \approx 0.05$–0.12 (e.g. ch 10 TIM – ch 17 VF)
- 5G NR (ch 25–28): near-stationary, low amplitude; $f_{ii} \approx 0.98$, off-diag ≈ 0
- Same-operator intra-band: $f_{ij} \in [0.05, 0.10]$; cross-operator inter-band: $|f_{ij}| \leq 0.05$; high $\hat{o}_{m,h} \Rightarrow$ reduce $\rho_{\text{rel},m}$ toward ϵ_ρ

Environment Details:

- Site: Univ. Hospital Tor Vergata, Rome; July 2023–March 2024; 7.5-min grid
- $M = 29$ channels, 9 kHz–6 GHz; $L = 1344$ steps (7 days); $H = 192$ steps (1 day)
- Operators: Iliad (ch 0–6), TIM (ch 7–13), VF (ch 14–20), W3 (ch 21–28)
- Bands: 2G (ch 0–7), 3G (ch 8–14), 4G (ch 15–24), 5G (ch 25–28)
- Kalman: $R_0 = 2.0$, $q = 10^{-3}$, $\varepsilon_\rho = 0.1$

Historical Context $\mathbf{X}_{\text{past}}$:

`[channel_stats_table]` *(ch_id | operator | band | $\hat{\mu}$ | $\hat{\sigma}$ | trend slope | $\text{autocorr}_{\text{lag-1}}$)*

TSFM Output θ (Chronos-2):

`[tsfm_forecast]`

Few-Shot Examples $\mathcal{S}_M$ (3 nearest from $\mathcal{D}$):

`[retrieved_examples]` *(context summary → $\mathbf{F}$ diag. range → ND/MAPE)*

Instructions:

Think step by step before producing the output:

1. Analyze $\mathbf{X}_{\text{past}}$ per channel: identify trend, periodicity, and $\text{autocorr}_{\text{lag-1}}$
2. Inspect $\boldsymbol{\theta}$: note which channels have large $\hat{o}_{m,h}$ and assign low $\rho_{\text{rel},m}^{(0)}$
3. Use domain knowledge and $\mathcal{S}_M$ to set diagonal $f_{ii}^{(0)}$ per band
4. Identify dominant cross-channel pairs and set off-diagonal $f_{ij}^{(0)}$ accordingly
5. Justify $\boldsymbol{\delta}_0^{(0)}$ from boundary dynamics at $t = 0$
6. Verify all eigenvalues of $\mathbf{F}^{(0)}$ satisfy $|\lambda| \leq 1$ before outputting

Response Format:

1. `<REASONING_TRACE>`: Follow the steps above explicitly; identify dominant channel pairs; justify $\boldsymbol{\delta}_0$ from boundary dynamics;
2. `<OUTPUT>`: Full JSON

Required Output Format:

```
{"F_coefficients":[{"i":int,"j":int,"f_ij":float},...],
 "delta_0":[float x29], "rho_rel":[float x29]}
```

Followed by `<REASONING_TRACE>`.

Output Block of LLM_1 - $k = 0$

Output: $\phi^{(0)} = \{\mathbf{F}^{(0)}, \boldsymbol{\delta}_0^{(0)}, \boldsymbol{\rho}_{\text{rel}}^{(0)}\}$

```
{"F_coefficients":[
   {"i":0, "j":0,  "f_ij":[f_2G_self]},
   {"i":1, "j":0,  "f_ij":[f_2G_intra]},
   {"i":10,"j":15, "f_ij":[f_3G4G_cross]},
   ... (all 29x29 entries) ],
 "delta_0":[[d_ch0],[d_ch1],...,[d_ch28]],
 "rho_rel":[[r_ch0],[r_ch1],...,[r_ch28]]}
```

Reasoning Trace $\text{RT}^{(0)}$:

LLM_1 - Full $\mathbf{F}^{(0)} \in \mathbb{R}^{29\times 29}$ populated as follows. Diagonal: $f_{ii}^{(0)} \in [0.95, 0.99]$ for 2G (ch 0–7) based on high $\text{autocorr}_{\text{lag-1}}$ in $\mathbf{X}_{\text{past}}$; $f_{ii}^{(0)} \in [0.90, 0.96]$ for 3G–4G (ch 8–24); $f_{ii}^{(0)} \approx 0.98$ for near-flat 5G (ch 25–28). Off-diagonal: cross-operator 3G–4G pairs (e.g. ch 10–ch 17, TIM–VF) assigned $f_{ij}^{(0)} \in [0.05, 0.12]$ cross-operator pairs $|f_{ij}^{(0)}| \leq 0.05$; remaining entries near zero. Per-channel $\rho_{\text{rel},m}^{(0)}$: channels with large uncertainty (e.g. ch 17, ch 21) assigned low $\rho_{\text{rel},m}^{(0)} \in [\epsilon_\rho, 0.3]$; channels with small uncertainty (e.g. ch 25–28) assigned $\rho_{\text{rel},m}^{(0)} \in [0.7, 1.0]$. All entries specified.

$k = 0$, Prompt Block $\Pi_E^{(0)}$ - Evaluator Agent

Task Description $\mathcal{T}_E$:

You are **LLM4TSF-Evaluator**. Analyze the submitted state-space parameters for plausibility and consistency with $\mathbf{X}_{\text{past}}$ and TSFM uncertainty $\hat{\mathbf{o}}$. Do **not** produce a forecast. Output *only* $\Delta^{(1)}$: *"No major changes needed"* or at most 5 lines of targeted revision instructions.

Environment Details:

(Same as Reasoning Agent) $M = 29$; Iliad/TIM/VF/W3; 2G/3G/4G/5G; $L = 1344$, $H = 192$, $R_0 = 2.0$, $q = 10^{-3}$

Domain Knowledge:

(Same as Reasoning Agent)

TSFM Output θ:

`[tsfm_forecast]`

Historical Context $\mathbf{X}_{\text{past}}$:

`[channel_stats_table]`

LLM Reasoning Trace $\text{RT}^{(0)}$:

`[reasoning_trace_k0]`

Submitted $\phi^{(0)}$:

$\mathbf{F}$: `[F_submitted]` $\boldsymbol{\delta}_0$: `[delta0_sub]` $\boldsymbol{\rho}_{\text{rel}}$: `[rho_sub]`

Few-Shot Examples $\mathcal{S}_E$:

`[eval_examples]` *(F submitted → issue detected → Δ given → MAPE gain)*

Chain-of-Thought Evaluation Instructions:

Think step by step before producing feedback:

1. Check diagonal entries $f_{ii}^{(0)}$ against band-level $\text{autocorr}_{\text{lag-1}}$ in $\mathbf{X}_{\text{past}}$; flag any that violate domain knowledge ranges
2. Verify off-diagonal entries $f_{ij}^{(0)}$ are consistent with operator and band coupling rules; flag entries outside expected ranges
3. Assess $\boldsymbol{\rho}_{\text{rel}}^{(0)}$–$\hat{o}_{m,h}$ inverse alignment: channels with large QS_m should have low $\rho_{\text{rel},m}^{(0)}$; flag violations
4. Check $\boldsymbol{\delta}_0^{(0)}$ is appropriately scaled relative to historical channel variations in $\mathbf{X}_{\text{past}}$
5. Inspect $\text{RT}^{(0)}$ for consistency between stated reasoning and submitted values
6. Summarize only flagged issues as $\Delta^{(1)}$; if none, output *"No major changes needed"*

Response Format:

1. `<FEEDBACK>`: Output $\Delta^{(1)}$ only

Output Block of LLM$_E$ - $k = 0$

Output Feedback:
$\Delta^{(1)}$: "Overall $\mathbf{F}^{(0)} \in \mathbb{R}^{29\times29}$ structure is plausible. Two per-channel issues detected. (1) Off-diagonal $f^{(0)}_{10,17}$ (ch 10–ch 17, 3G–4G, TIM) exceeds the expected range $[0.05, 0.12]$ for moderate cross-band coupling – reduce to within $[0.05, 0.12]$. (2) $\rho^{(0)}_{\text{rel},17}$ for ch 17 is too high, reduce $\rho^{(0)}_{\text{rel},17}$ to shift Kalman trust toward LLM-defined dynamics for this channel. All other entries consistent."

$k = 1$, Prompt Block $\Pi^{(1)}$ - LLM Reasoning Agent

Evaluator Feedback $\Delta^{(1)}$ [*address every point first*]:
`[delta_1_full]`

Previous Output $\phi^{(0)}$ and History:
$\mathcal{H}^{(1)} = \mathcal{H}^{(0)} \cup \{\Delta^{(1)}, \phi^{(0)}, \text{RT}^{(0)}\}$; $\mathcal{H}^{(0)} = \{\mathbf{X}_{\text{past}}, \theta, \mathcal{S}_M, \mathcal{T}\}$
$\mathbf{F}$: `[F_k0]` $\boldsymbol{\delta}_0$: `[d0k0]` $\boldsymbol{\rho}_{\text{rel}}$: `[rhok0]`

Previous Reasoning Trace RT$^{(0)}$:
`[reasoning_trace_k0]`

Response Format:

1. `<REASONING_TRACE>`: For each evaluator flag - state what was wrong, how it is corrected, and why the correction is consistent with EMF domain knowledge and θ.
2. `<OUTPUT>`: Full revised JSON (all 29×29 entries; do not omit unchanged entries)

Required Output Format:

```
{"F_coefficients":[{"i":int,"j":int,"f_ij":float},...],
 "delta_0":[float x29], "rho_rel":[float x29]}
```

Output Block of LLM$_1$ - $k = 1$

Output: $\phi^{(1)} = \{\mathbf{F}^{(1)}, \boldsymbol{\delta}^{(1)}_0, \boldsymbol{\rho}^{(1)}_{\text{rel}}\}$

```
{"F_coefficients":[
   {"i":0, "j":0,  "f_ij":[f_2G_self]},       % unchanged
   {"i":10,"j":17, "f_ij":[f_3G4G_revised]},   % corrected
   ... (all 29x29 entries) ],
 "delta_0":[[d_ch0],...,[d_ch28]],              % unchanged
 "rho_rel":[[r_ch0],...,[r_revised],...,[r_ch28]]}
```

Reasoning Trace RT$^{(1)}$:
LLM$_1$ - Applied $\Delta^{(1)}$ to $\mathbf{F}^{(0)} \in \mathbb{R}^{29\times29}$: (1) reduced $f^{(1)}_{10,17}$ to within $[0.05, 0.12]$, consistent with moderate 3G–4G cross-operator coupling (ch 10, TIM; ch 17, VF) observed in $\mathbf{X}_{\text{past}}$. (2) reduced $\rho^{(1)}_{\text{rel},17}$ toward ϵ_ρ, consistent with ch 17 exhibiting one of the largest uncertainty values across all 29 channels. All remaining entries of $\mathbf{F}^{(1)}$, $\boldsymbol{\delta}^{(1)}_0$, and $\boldsymbol{\rho}^{(1)}_{\text{rel}}$ carried unchanged from $\boldsymbol{\phi}^{(0)}$.

$k = 1$, Prompt Block $\Pi^{(1)}_E$ - Evaluator Agent

This is the evaluation round $k = 1$. Determine whether $\phi^{(1)}$ correctly incorporates your prior feedback $\Delta^{(1)}$. If all issues are resolved output *"No major changes needed. Proceed to Kalman fusion."* Otherwise output residual corrections as $\Delta^{(2)}$ (max 5 lines).

Required Output Format:
$\Delta^{(2)}$: `<feedback or PASS>`

Prior Feedback $\Delta^{(1)}$:
`[delta_1_full]`

Revised Submission $\phi^{(1)}$ (under review):
$\mathbf{F}$: `[Fk1]` $\boldsymbol{\delta}_0$: `[d0k1]` $\boldsymbol{\rho}_{\text{rel}}$: `[rhok1]`

LLM Reasoning Trace RT$^{(1)}$:
`[reasoning_trace_k1]`

Response Format:

1. `<FEEDBACK>`: Output $\Delta^{(2)}$ only

Output Block of LLM$_E$ - $k = 1$

Output Feedback:
$\Delta^{(2)}$: "No major changes needed. Proceed to Kalman fusion. Revised $f^{(1)}_{10,17}$ now within $[0.05, 0.12]$, consistent with 3G–4G working-hour coupling in $\mathbf{X}_{\text{past}}$ (TIM, ch 10 and ch 17). Updated $\rho^{(1)}_{\text{rel},17}$ toward ϵ_ρ correctly reflects the large uncertainty for ch 17. All entries of $\mathbf{F}^{(1)}$ verified; all $\boldsymbol{\delta}^{(1)}_0$ entries consistent with $\mathbf{X}_{\text{past}}$ boundary conditions. $\boldsymbol{\phi}^{(1)}$ accepted."

Output Parsing and Validation

The parser extracts the JSON content enclosed by the `<OUTPUT>` tags. The fields `F_coefficients`, `delta_0`, and `rho_rel` are converted into $\mathbf{F}^{(k)}$, $\boldsymbol{\delta}^{(k)}_0$, and $\boldsymbol{\rho}^{(k)}_{\text{rel}}$, respectively. It verifies that $\mathbf{F}^{(k)}$ contains M^2 finite numerical entries with valid channel indices, both vectors contain M finite entries, and each reliability value lies in $[\epsilon_\rho, 1]$. Outputs with missing fields, invalid dimensions, or non-numerical values are rejected and regenerated.

Prompt-Sensitivity Analysis

Table 1 reports prompt-sensitivity analysis, where Prompt 1 contains a basic task description, Prompt 2 additionally includes few-shot examples $\mathcal{S}_M$, and Prompt 3 further adds chain-of-thought reasoning guidance; all three prompt versions are shown on the next page of this document. Prompt 3, which provides the most comprehensive reasoning context through domain knowledge, forecasting objectives, and explicit reasoning guidance, consistently achieves the best forecasting accuracy across all technology bands. The largest gains are observed on 2G and 3G, where MAPE decreases from 18.16 to 14.69 and from 16.45 to 12.14, respectively. These results indicate that combining few-shot examples with explicit reasoning guidance enables the LLM to generate more effective dependency and reliability estimates for Kalman fusion.

Table 1: Forecasting performance across three prompt configurations. Lower is better.

Prompt Style	Band	ND	NRMSE	MAPE
Prompt 1	2G	0.15313	0.1735	18.1587
	3G	0.1566	0.1898	16.4532
	4G	0.1174	0.1471	12.6664
	5G	0.1097	0.1598	10.2403
Prompt 2	2G	0.1414	0.1664	14.8712
	3G	0.1326	0.1653	15.5841
	4G	0.1147	0.1469	11.9985
	5G	0.0964	0.1549	9.1948
Prompt 3	2G	**0.1323**	**0.1520**	**14.6899**
	3G	**0.1047**	**0.1319**	**12.1411**
	4G	**0.1122**	**0.1367**	**11.4345**
	5G	**0.0849**	**0.1309**	**8.1475**

LLM4TSF: Prompt Configurations for Prompt-Sensitivity Analysis

Three prompt used in Table IV of the main paper, **Highlighted sections** indicate additions over the previous prompt.

Prompt 1 - Task Description Only

Task Description $\mathcal{T}$:

You are **LLM4TSF-Reasoner**, an expert in multivariate RF-EMF analysis and cross-channel state-space modelling. Given $\mathbf{X}_{\text{past}}$ and TSFM output $\boldsymbol{\theta}$, produce the state-transition specification $\{\mathbf{F}^{(0)}, \boldsymbol{\delta}_0^{(0)}, \boldsymbol{\rho}_{\text{rel}}^{(0)}\}$ for the Kalman fusion layer.

Objective - generate parameters that:

- Capture cross-channel EMF dependencies via $\mathbf{F} \in \mathbb{R}^{29\times 29}$;
- Provide initial-state corrections $\boldsymbol{\delta}_0 \in \mathbb{R}^{29}$ from boundary dynamics
- Assign reliability weights $\boldsymbol{\rho}_{\text{rel}} \in [\epsilon_\rho, 1]^{29}$: high $\hat{o}_{m,h} \Rightarrow$ low $\rho_{\text{rel},m}$

Domain Knowledge:

- 2G (ch 0–7): autocorr$_{\text{lag-1}} \geq 0.92$; set $f_{ii} \in [0.95, 0.99]$; strong daily periodicity
- 3G–4G cross-operator: moderate working-hour coupling (08:00–20:00 Rome); off-diag $f_{ij} \approx 0.05$–0.12 (e.g. ch 10 TIM – ch 17 VF)
- 5G NR (ch 25–28): near-stationary, low amplitude; $f_{ii} \approx 0.98$, off-diag ≈ 0
- Same-operator intra-band: $f_{ij} \in [0.05, 0.10]$; cross-operator inter-band: $|f_{ij}| \leq 0.05$; high $\hat{o}_{m,h} \Rightarrow$ reduce $\rho_{\text{rel},m}$ toward ϵ_ρ

Environment Details:

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- Operators: Iliad (ch 0–6), TIM (ch 7–13), VF (ch 14–20), W3 (ch 21–28)
- Bands: 2G (ch 0–7), 3G (ch 8–14), 4G (ch 15–24), 5G (ch 25–28)
- Kalman: $R_0 = 2.0$, $q = 10^{-3}$, $\varepsilon_\rho = 0.1$

Historical Context $\mathbf{X}_{\text{past}}$:

`[channel_stats_table]` *(ch_id | operator | band | $\hat{\mu}$ | $\hat{\sigma}$ | trend slope | autocorr$_{\text{lag-1}}$)*

TSFM Output $\boldsymbol{\theta}$ (Chronos-2):

`[tsfm_forecast]`

Response Format:

1. `<OUTPUT>`: Full JSON

Required Output Format:

```
{"F_coefficients":[{"i":int,"j":int,"f_ij":float},...],
 "delta_0":[float x29], "rho_rel":[float x29]}
```

Followed by `<REASONING_TRACE>`.

Prompt 2 - Task Description + Few-Shot Examples $\mathcal{S}_M$

Task Description $\mathcal{T}$:

You are **LLM4TSF-Reasoner**, an expert in multivariate RF-EMF analysis and cross-channel state-space modelling. Given $\mathbf{X}_{\text{past}}$ and TSFM output $\boldsymbol{\theta}$, produce the state-transition specification $\{\mathbf{F}^{(0)}, \boldsymbol{\delta}_0^{(0)}, \boldsymbol{\rho}_{\text{rel}}^{(0)}\}$ for the Kalman fusion layer.

Objective - generate parameters that:

- Capture cross-channel EMF dependencies via $\mathbf{F} \in \mathbb{R}^{29\times 29}$;
- Provide initial-state corrections $\boldsymbol{\delta}_0 \in \mathbb{R}^{29}$ from boundary dynamics
- Assign reliability weights $\boldsymbol{\rho}_{\text{rel}} \in [0, 1]^{29}$: high $\hat{o}_{m,h} \Rightarrow$ low $\rho_{\text{rel},m}$

Domain Knowledge:

- 2G (ch 0–7): autocorr$_{\text{lag-1}} \geq 0.92$; set $f_{ii} \in [0.95, 0.99]$; strong daily periodicity
- 3G–4G cross-operator: moderate working-hour coupling (08:00–20:00 Rome); off-diag $f_{ij} \approx 0.05$–0.12 (e.g. ch 10 TIM – ch 17 VF)
- 5G NR (ch 25–28): near-stationary, low amplitude; $f_{ii} \approx 0.98$, off-diag ≈ 0
- Same-operator intra-band: $f_{ij} \in [0.05, 0.10]$; cross-operator inter-band: $|f_{ij}| \leq 0.05$; high $\hat{o}_{m,h} \Rightarrow$ reduce $\rho_{\text{rel},m}$ toward ϵ_ρ

Environment Details:

- Site: Univ. Hospital Tor Vergata, Rome; July 2023–March 2024; 7.5-min grid
- $M = 29$ channels, 9 kHz–6 GHz; $L = 1344$ steps (7 days); $H = 192$ steps (1 day)
- Operators: Iliad (ch 0–6), TIM (ch 7–13), VF (ch 14–20), W3 (ch 21–28)
- Bands: 2G (ch 0–7), 3G (ch 8–14), 4G (ch 15–24), 5G (ch 25–28)
- Kalman: $R_0 = 2.0$, $q = 10^{-3}$, $\varepsilon_\rho = 0.1$

Historical Context $\mathbf{X}_{\text{past}}$:

`[channel_stats_table]` *(ch_id | operator | band | $\hat{\mu}$ | $\hat{\sigma}$ | trend slope | autocorr$_{\text{lag-1}}$)*

TSFM Output $\boldsymbol{\theta}$ (Chronos-2):

`[tsfm_forecast]`

$\oplus$ **Few-Shot Examples $\mathcal{S}_M$ (3 nearest from $\mathcal{D}$):**

`[retrieved_examples]` *(context summary → $\mathbf{F}$ diag. range → ND/MAPE)*

Response Format:

1. `<OUTPUT>`: Full JSON

Required Output Format:

```
{"F_coefficients":[{"i":int,"j":int,"f_ij":float},...],
 "delta_0":[float x29], "rho_rel":[float x29]}
```

Followed by `<REASONING_TRACE>`.

Prompt 3 - Task Description + Few-Shot Examples $\mathcal{S}_M$ + Chain-of-Thought

Task Description $\mathcal{T}$:

You are **LLM4TSF-Reasoner**, an expert in multivariate RF-EMF analysis and cross-channel state-space modelling. Given $\mathbf{X}_{\text{past}}$ and TSFM output $\boldsymbol{\theta}$, produce the state-transition specification $\{\mathbf{F}^{(0)}, \boldsymbol{\delta}_0^{(0)}, \boldsymbol{\rho}_{\text{rel}}^{(0)}\}$ for the Kalman fusion layer.

Objective - generate parameters that:

- Capture cross-channel EMF dependencies via $\mathbf{F} \in \mathbb{R}^{29\times 29}$;
- Provide initial-state corrections $\boldsymbol{\delta}_0 \in \mathbb{R}^{29}$ from boundary dynamics
- Assign reliability weights $\boldsymbol{\rho}_{\text{rel}} \in [0, 1]^{29}$: high $\hat{o}_{m,h} \Rightarrow$ low $\rho_{\text{rel},m}$

Domain Knowledge:

- 2G (ch 0–7): autocorr$_{\text{lag-1}} \geq 0.92$; set $f_{ii} \in [0.95, 0.99]$; strong daily periodicity
- 3G–4G cross-operator: moderate working-hour coupling (08:00–20:00 Rome); off-diag $f_{ij} \approx 0.05$–0.12 (e.g. ch 10 TIM – ch 17 VF)
- 5G NR (ch 25–28): near-stationary, low amplitude; $f_{ii} \approx 0.98$, off-diag ≈ 0
- Same-operator intra-band: $f_{ij} \in [0.05, 0.10]$; cross-operator inter-band: $|f_{ij}| \leq 0.05$; high $\hat{o}_{m,h} \Rightarrow$ reduce $\rho_{\text{rel},m}$ toward ϵ_ρ

Environment Details:

- Site: Univ. Hospital Tor Vergata, Rome; July 2023–March 2024; 7.5-min grid
- $M = 29$ channels, 9 kHz–6 GHz; $L = 1344$ steps (7 days); $H = 192$ steps (1 day)
- Operators: Iliad (ch 0–6), TIM (ch 7–13), VF (ch 14–20), W3 (ch 21–28)
- Bands: 2G (ch 0–7), 3G (ch 8–14), 4G (ch 15–24), 5G (ch 25–28)
- Kalman: $R_0 = 2.0$, $q = 10^{-3}$, $\varepsilon_\rho = 0.1$

Historical Context $\mathbf{X}_{\text{past}}$:

`[channel_stats_table]` *(ch_id | operator | band | $\hat{\mu}$ | $\hat{\sigma}$ | trend slope | autocorr$_{\text{lag-1}}$)*

TSFM Output $\boldsymbol{\theta}$ (Chronos-2):

`[tsfm_forecast]`

Few-Shot Examples $\mathcal{S}_M$ (3 nearest from $\mathcal{D}$):

`[retrieved_examples]` *(context summary → $\mathbf{F}$ diag. range → ND/MAPE)*

⊕ **Chain-of-Thought Reasoning Instructions:**

Think step by step before producing the output:

1. Analyze $\mathbf{X}_{\text{past}}$ per channel: identify trend, periodicity, and $\text{autocorr}_{\text{lag-1}}$
2. Inspect $\boldsymbol{\theta}$: note which channels have large $\hat{o}_{m,h}$ and assign low $\rho^{(0)}_{\text{rel},m}$
3. Use domain knowledge and $\mathcal{S}_M$ to set diagonal $f^{(0)}_{ii}$ per band
4. Identify dominant cross-channel pairs and set off-diagonal $f^{(0)}_{ij}$ accordingly
5. Justify $\boldsymbol{\delta}^{(0)}_0$ from boundary dynamics at $t = 0$
6. Verify all eigenvalues of $\mathbf{F}^{(0)}$ satisfy $|\lambda| \leq 1$ before outputting

Response Format:

1. `<REASONING_TRACE>`: Follow the steps above explicitly; identify dominant channel pairs; justify $\boldsymbol{\delta}_0$ from boundary dynamics;
2. `<OUTPUT>`: Full JSON

Required Output Format:

```
{"F_coefficients":[{"i":int,"j":int,"f_ij":float},...],
 "delta_0":[float x29], "rho_rel":[float x29]}
```

Followed by `<REASONING_TRACE>`.

Baseline Configurations

The trainable baselines use the first 60% of each dataset for training, the next 20% for validation, and the final 20% for testing. All methods use the same lookback length L and forecast horizon H as LLM4TSF. The neural baselines use MSE loss, the Adam optimizer, a batch size of 32, and early stopping. Per-channel standardization is performed using training-set statistics only, and the random seed is fixed to 42.

Table 2: Configurations of the forecasting baselines.

Method	Main configuration	LR	Epochs	Patience
VAR	Lag order 12	–	–	–
ARIMA	Per-channel order $(5, 0, 0)$	–	–	–
LSTM	64 units; dropout 0.25	10^{-3}	50	10
XGBoost	100 trees; depth 6; subsample 0.8	0.05	–	–
DLinear	Per-channel; kernel size 25	10^{-3}	100	10
PatchTST	Per-channel; patch 16; stride 8; $d = 64$; 4 heads; 2 layers; dropout 0.1	10^{-4}	100	15
Chronos-2	Zero-shot; no task-specific training	–	–	–